Math 142, F19 **Name:** Answer

Quiz 7

<u>**Instructions:**</u> **You must show all of your work to receive credits for correct answers.**

Problem 1. (10 points) Determine whether the sequence $\{a_n\}_{n=1}^{+\infty}$ converges or diverges and find the limit if it converges.

(a) $a_n = \dfrac{2+5n^2}{n+n^2}$

$$\lim_{n\to+\infty} a_n = \lim_{n\to+\infty} \frac{2+5n^2}{n+n^2}$$

$$= \lim_{n\to+\infty} \frac{(2+5n^2)/n^2}{(n+n^2)/n^2}$$

$$= \lim_{n\to+\infty} \frac{\frac{2}{n^2}+5}{\frac{1}{n}+1}$$

$$= \frac{5}{1} = 5$$

Convergent with $L=5$

(b) $a_n = \dfrac{2\sin(n)}{n}$ (*Hint: using Squeezing Thm.*)

$$-\frac{2}{n} \leq \frac{2\sin n}{n} \leq \frac{2}{n}$$

let $n \to +\infty$

$$0 \leq \lim_{n\to+\infty} \frac{2\sin n}{n} \leq 0$$

$$\Rightarrow \lim_{n\to+\infty} a_n = 0$$

Convergent with $L=0$

(c) $a_n = (1+\frac{3}{n})^n$

$= e^3$ convegent

Since $\lim_{n\to+\infty} (1+\frac{x}{n})^n = e^x$

(d) $a_n = \dfrac{n^2}{e^n}$ (*Hint: using L'Hopital's rule*)

let $f(x) = \dfrac{x^2}{e^x}$ then $a_n = f(n)$

$$\lim_{x\to+\infty} f(x) = \lim_{x\to+\infty} \frac{x^2}{e^x}$$

$$= \lim_{x\to+\infty} \frac{2x}{e^x} = \lim_{x\to+\infty} \frac{2}{e^x}$$

$$= 0$$

thus $\lim_{n\to+\infty} a_n = 0$

convergent with $L=5$